It is recommended that you spend about 30 minutes on this question.

2 Beta particles can be deflected by magnetic fields.

Design a laboratory experiment to investigate how magnetic flux density affects the angle through which beta particles are deflected when they pass through a uniform magnetic field. The only radioactive source that is available to you is a Radium-226 source which emits α, β and γ radiation.

You should draw a diagram showing the arrangement of your apparatus. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the method by which beta particles **only** would be detected,

(c) the method of measuring the angle of deflection,

(d) how the magnetic field would be produced, measured and changed,

(e) any safety precautions you would take.

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